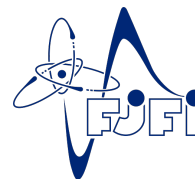




02YMECH/02YMECHZ Mechanics



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Course Description

02YMECH is a winter semester course for first-year bachelor students covering the basics of mechanics and some advanced methods, such as solving equations of motion for mechanical systems like a particle under constant force, a harmonic oscillator, the central force problem, and rigid bodies.

Each week, there will be four 50-minute lectures and two 50-minute exercise session. Lectures will cover theoretical concepts following the weekly schedule, while exercise sessions will focus on solving problems related to the lecture material. The instruction language is English.

02YMECHZ refers to the final exam for 02YMECH. The students who pass 02YMECH will be eligible to take 02YMECHZ. Please see the exam section for more details.

Prerequisite(s)

Knowledge of basic differentiation, integration, and trigonometry would be useful but are not necessary.

Credits: 4 (02YMECH) + 2 (02YMECHZ)

Reference Textbooks

1. Morin, D. J., *Introduction to Classical Mechanics: With Problems and Solutions*, Cambridge University Press, 2008.
2. Taylor, J. R., *Classical Mechanics*, University Science Books, 2013.
3. Thornton, S. T., Marion, J. B., - *Classical Dynamics of Particles and Systems*, 3rd Edition, Brooks/Cole, 2003.
4. Halliday, D., Resnick, R., Walker, J., *Fundamentals of Physics*, 10th Edition (Extended), Wiley, 2013.

Course Objectives

1. To learn the basics of mechanics, and solve simple equations of motion.
2. To understand the general framework for constructing and operating physical models.

Exams

There will be **two midterm exams** and **one final exam** (02YMECHZ). Each exam will consist of **4 questions**, with each question worth **1 point**.

- Students who score at least **4 points out of 8** across the two midterm exams will be awarded the **Z (Zápočet)** grade. This means that they have passed 02YMECH, received 4 credits, and are eligible to take the final exam.
- Students who are not awarded the Z grade but score **at least 1 point in total** across the two midterm exams may take a makeup exam upon request. The makeup exam will cover the same material as the 2nd Midterm Exam. To obtain the Z grade through the makeup exam, students must score **at least 5 points in total** across the two midterm exams and the makeup exam. Students who score less than 1 point across the two midterms, or who do not reach the required 5-point total after the makeup exam, must retake the course.
- The final exam consists of a **written component** and an **oral component**.
- Only students who score at least **2 points on the written part** may proceed to the oral examination. Students who score less than 2 points on the written part fail 02YMECHZ, and an **F grade** will be registered.
- Attendance at the oral examination is **mandatory** for students who pass the written part.
- The oral examination will be conducted as follows: Each student will randomly select a number from 1 to 16, corresponding to one of the sixteen topics listed at the end of this syllabus (see *Topics for the Oral Part of the Final Exam*). Students will then have 30 minutes to prepare the chosen topic on blank paper, followed by a 5–10-minute oral examination based on that topic. Students are not required to provide all derivations unless otherwise stated, but they should be able to explain the main concepts and equations related to the selected topic.
- The **final letter grade** is primarily determined by the written exam score. However, if a student demonstrates an outstanding performance during the oral examination, the final letter grade will be increased by one level. Conversely, if a student demonstrates a lack of understanding of the basic concepts of the course, the final letter grade will be decreased by one level. An average performance during the oral examination will not affect the final letter grade.
- On request, the lecturer may allow a student to retake the final exam before the beginning of the next academic semester, **provided that the student has scored at least 1.5 points on the final exam**. Otherwise, the student must take the final exam in the next examination period.
- Students who have already been awarded the Z grade but fail the final exam do not need to take the midterm exams again in the following academic year.

The exam dates will be announced in KOS during the second week of the semester. Students are responsible for registering for the exams in KOS. Students are required to bring their student ID cards to the exams.

Homework

Students can find homework problems in the course repository, associated with each chapter. The purpose of the homework is to allow students to work independently and familiarize themselves with the subject. Homework will not be collected or graded.

Letter Grade Distribution

$[3.6 - 4.0]$: A = Excellent	$[2.4 - 2.8]$: D = Average
$[3.2 - 3.6]$: B = Very Good	$[2.0 - 2.4]$: E = Sufficient
$[2.8 - 3.2]$: C = Good	$[0.0 - 2.0]$: F = Insufficient

Attendance and Absences

- Attendance is strongly recommended but not mandatory. *Based on the lecturer's experience over the years, regular attendance is strongly associated with successful completion of the course.*
- Students are responsible for all material covered during any classes they miss, regardless of the reason for their absence. Students are also responsible for obtaining any missed lecture notes, materials, or other information.

Tentative Course Outline

The weekly coverage might change as it depends on the progress of the class.

Chapter	Week
Mathematical preliminaries	1
Kinematics	2
Dynamics	3
Rotation	4
Conservation theorems	5,6
Small oscillations	7,8
Central-force motion	9,10
Dynamics of system of particles	11
Motion in a non-inertial reference frame	12
Dynamics of rigid bodies	13

Week	Content
Week 1	Scalars, scalar functions, cartesian and polar coordinates, derivatives, integrals, introduction to first- and second-order homogenous differential equations, vectors, vector operations.
Week 2	Reference frames, position, displacement, velocity, acceleration, speed, average speed, momentum, basic types of motion and their superposition.
Week 3	Inertial and non-inertial reference frames, Newton's laws of motion, equation of motion for a particle under a constant force, inclined plane, tension, drag force, friction, freefall and terminal speed, projectile motion.
Week 4	Angular velocity, torque, angular momentum, rotational inertia of a system of particles, Newton's 2nd law in rotational form, uniform circular motion.
Week 5	Conservation of linear and angular momentum, work and kinetic energy, conservative forces and potential energy.
Week 6	Conservation of total energy, total energy as an equation of motion, equilibrium, impulse, power.
Week 7	Simple harmonic oscillator, simple pendulum, phase diagrams for the oscillative behavior.
Week 8	Damped oscillations, driven oscillations, resonance.
Week 9	Definition of central force and its properties, equations of motion for central force problem, orbits, centrifugal energy, effective potential.
Week 10	Two-body problem, planetary motion (cyclic, elliptic, parabolic, and hyperbolic orbits, eccentricity, and total energy), Kepler's laws of planetary motion.
Week 11	Center of mass, reconsideration of basic mechanical quantities, dynamics and conservation laws for system of particles, elastic and inelastic collisions.
Week 12	Centrifugal and coriolis forces, motion relative to the Earth, oblateness of Earth, cyclonic motion in atmosphere.
Week 13	Yo-yo problem, physical pendulum, inertia tensor, Euler's equations, force-free motion of a symmetric top.

Topics for the Oral Part of the Final Exam

1. How would you describe a circular motion in terms of a complex number? If you wanted the radius of the circular trajectory to decrease exponentially with time, how would you modify your description? What would be the connection of this description to damped oscillations?
2. Derive work-energy theorem step-by-step starting from the definition of work and Newton's second law and explain its meaning.
3. Define conservative force and potential energy by using the definition of work and explain why a conservative force can be expressed as the negative gradient of potential energy.
4. Show mathematically step-by-step in detail why the total mechanical energy is conserved for a particle moving under a conservative force.
5. Use the conservation of mechanical energy as an equation of motion to derive the time dependence of position for a simple harmonic oscillator in one dimension.
6. Derive step-by-step of total kinetic energy for discrete and continuous mass distributions starting from the definition of kinetic energy for a single particle and describe of how we define moment of inertia on the way.
7. Describe projectile motion and derive expressions for time of flight, maximum height, range and trajectory equation.
8. If the potential energy of a mechanical system can be described as an analytical function and if we are interested in small oscillations around the equilibrium point, show step-by-step how we can approximate the potential energy function using Taylor series expansion and derive the equation of motion in differential equation for for small oscillations. Do not solve it but memorize the general solution in terms of trigonometric functions.
9. Starting from Hooke's law, derive the equation of motion for a simple harmonic oscillator and solve it step-by-step to get the general solution. Explain the physical meaning of each term in the solution.
10. Starting from the expression $x(t) = x_0 \cos(\omega_0 t - \delta)$, show the form of the trajectories followed by a simple harmonic oscillator in phase space (the phase diagram). Describe all relevant features of the phase diagram in terms of physical parameters of the system, such as total energy, amplitude, frequency, and mass. How would the phase diagram change if we introduce damping into the system?
11. Write down the equation of motion for a damped harmonic oscillator under the influence of a damping force proportional to velocity. Solve the equation step-by-step and obtain the general solution in exponential form. Explain briefly how different types of damping based on the relationship between the damping coefficient and the natural frequency of the system lead to different forms of the motion (underdamping, overdamping, and critical damping).
12. Write down and solve step-by-step the equation of motion for a driven damped harmonic oscillator under the influence of an external sinusoidal driving force to derive the expression for the amplitude and phase difference of steady-state oscillations. Explain the concept of resonance and how it manifests in this system.

13. Define central force and describe its three main properties: No torque, motion in a plane, equal areas in equal times. Show mathematically step-by-step how each property can be derived from the definition of central force.
14. Describe how two-body problem can be reduced to an equivalent one-body problem. Derive the equation of motion for the equivalent one-body problem step-by-step.
15. Write down the effective potential for inverse-square central force and draw its graph. Explain how different types of orbits (circular, elliptical, parabolic, hyperbolic) can be understood in terms of the effective potential and total energy of the system.
16. Write down the effective force in a non-inertial reference frame including all non-inertial terms. Explain the physical meaning of each term in detail. Describe the cyclonic motion in the atmosphere in terms of these non-inertial forces and explain why the wind blows counterclockwise (clockwise) around low-pressure areas in the northern (southern) hemisphere.